

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Active-Controlled, Double-Blind Clinical Study to Evaluate the Antiretroviral Activity, Safety, and Tolerability of Doravirine/Islatravir (DOR/ISL 100 mg/0.25 mg) Once-Daily in HIV-1 Infected Treatment-Naïve Participants **Short Title/Acronym:** DOR/ISL in HIV-1 Antiretroviral Treatment-naïve Participants (MK-8591A-053) **Protocol Number:** MK-8591A-053 **NCT Number:** NCT05705349 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Doravirine/Islatravir (DOR/ISL 100 mg/0.25 mg) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * Primary Objective: To evaluate the antiretroviral activity, safety, and tolerability of once-daily DOR/ISL (100 mg/0.25 mg) in treatment-naïve adults with HIV-1 infection compared to the once-daily 3-drug combination of BIC/FTC/TAF (Bictegravir/Emtricitabine/Tenofovir Alafenamide), demonstrating non-inferiority as assessed by the percentage of participants with HIV-1 RNA < 50 copies/mL at Week 48. *

Secondary Objectives: To evaluate the percentage of participants experiencing adverse events (AEs) and discontinuations due to AEs up to week 48, changes from baseline in CD4+ T-cell counts, and the development of viral drug resistance.

2.2 Background & Rationale As treatment regimens have improved, HIV-1 infection has become a chronic, manageable condition. The current standard of care includes 3-drug regimens (such as BIC/FTC/TAF). However, there is evidence that 2-drug regimens without an integrase inhibitor can be as effective while lowering pill burden, improving tolerability, and supporting long-term patient adherence to treatment. The management pathway for this study aligns with the standard clinical pathway for HIV-1 using Antiretroviral Therapy (ART). This study generates evidence to support DOR/ISL as a meaningful new, once-daily 2-drug treatment option for people with HIV-1 who have not previously taken antiretroviral therapies.

3. Study Design & Methodology

3.1 Design Framework * Study Type: Interventional *** Control Type:** Active Comparator (Bictegravir/Emtricitabine/Tenofovir Alafenamide - BIC/FTC/TAF) *** Blinding:** Double-blind (Participant, Investigator) *** Randomization:** 1:1

3.2 Planned Enrollment & Duration * **Target N\$:** 537 adult participants * **Number of Sites:** Global, multi-center trial (spanning multiple countries including the USA, Switzerland, and the UK) * **Estimated Study Start:** 08-Mar-2023 * **Estimated Primary Completion:** 19-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female ≥ 18 years of age. 2. Confirmed diagnosis of HIV-1 infection with an HIV viral load of ≥ 500 HIV-1 RNA copies/mL at the time of screening. 3. Treatment-naïve: Participants must not have started any prior treatment with antiretroviral (ARV) medications following their diagnosis.

4.2 Exclusion Criteria 1. Has HIV-2 infection. 2. Has active hepatitis B infection (defined as HBsAg-positive or HBV DNA-positive) or chronic hepatitis C virus (HCV) infection with laboratory values consistent with cirrhosis. 3. Diagnosis of an active AIDS-defining opportunistic infection within 30 days prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Doravirine/Islatravir (DOR/ISL 100 mg/0.25 mg) tablet taken orally once daily. * **Route of Administration:** Oral * **Duration of Treatment:** 144 follow-up weeks (with primary endpoint readout at Week 48 and secondary at Week 96).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard treatments for co-morbidities that do not present drug-drug interactions with DOR/ISL. * **Prohibited:** Other concurrent antiretroviral (ARV) therapies outside of the assigned study interventions, strong CYP3A inducers/inhibitors, or prior investigational HIV-1 medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -45 to 0	Informed Consent, Medical History, Baseline HIV-1 RNA/CD4+ Labs, Hepatitis screening
Baseline	Day 0	Randomization (1:1), First Dose Administration, Medication adherence counseling and safe practice education
Interim	Weeks 4 to 132	Onsite Visits at regular intervals (up to Weeks 48, 96, and 144): Safety Labs, CD4+ count, Efficacy Assessment (HIV-1 RNA), Medication Adherence (Checklist review every 12 weeks)
End of Study	Week 144	Final Vital Signs, Exit Interview, IP Accountability, Transition to open-label extension phase if applicable

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Percentage of participants achieving viral suppression (HIV-1 RNA < 50 copies/mL) at Week 48 (FDA snapshot approach for non-inferiority). * **Measurement Method:** Standardized central laboratory plasma HIV-1 RNA quantitative assay.

7.2 Secondary & Exploratory Endpoints * Percentage of participants with adverse events (AEs) and serious adverse events (SAEs) up to Week 48. * Percentage of participants discontinuing study intervention due to AEs up to Week 48. * Mean change from baseline in CD4+ T-cell count and the emergence of viral drug resistance.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection through participant reporting and continuous safety lab assessments (e.g., tracking lymphocyte/CD4+ counts).
 - **Serious Adverse Events (SAEs):** Captured throughout the treatment timeframe and reported typically within 24 hours of investigator awareness.
 - **Data Monitoring Committee (DMC):** External Data Monitoring Committee oversees unblinded safety and efficacy data throughout the 144-week period.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Full Analysis Set (FAS) / Intent-to-Treat (ITT) population for primary efficacy outcomes. Safety Set for AE/SAE evaluations.
 - **Sample Size Justification:** Approximately 537 randomized patients to provide appropriate power ($\alpha = 0.05$) to establish non-inferiority with a pre-specified margin (typically 10% for ARV trials).
 - **Handling of Missing Data:** US FDA Snapshot approach (missing data treated as failure/non-responder for viral suppression endpoints).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. All participating healthcare professionals (HCPs) will undergo an education protocol consisting of investigational ARV protocol training and double-blind masking adherence.

- **Monitoring Plan:** Standardized site visits and remote monitoring to verify protocol adherence, accurate AE recording, and drug accountability. The remote monitoring frequency includes continuous safety monitoring and AE tracking.
 - **Quality Audit Mechanism:** Centralized viral load monitoring and clinical site adherence tracking will be employed to ensure data integrity.
 - **Audit Trail:** eCRF locking, source data verification, and data clarification forms managed by Merck clinical data management.
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11. Study Identifiers & Governance

Primary ID: {{MK-8591A-053}} **Registry Number:** {{NCT05705349}} **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}} **Study Title:** {{DOR/ISL in HIV-1 Antiretroviral Treatment-naïve Participants}} **Official Regulatory Title:** {{A Phase 3, Randomized, Active-Controlled, Double-Blind Clinical Study to Evaluate the Antiretroviral Activity, Safety, and Tolerability of Doravirine/Islatravir (DOR/ISL 100 mg/0.25 mg) Once-Daily in HIV-1 Infected Treatment-Naïve Participants}}

12. Clinical Framework (HIV-1 Specific)

Primary Condition: {{HIV-1 Infection}} **Diagnosis Threshold:** {{HIV viral load ≥ 500 HIV-1 RNA copies/mL}} **Medical Methodology:** {{Antiretroviral Therapy (ART)}} **Optimization Target:** {{Viral suppression (HIV-1 RNA < 50 copies/mL)}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for HIV-1}} **Education Protocol (HCP):** {{Investigational ARV protocol training and double-blind masking adherence}} **Education Protocol (Patient):** {{Medication adherence counseling and safe practice education}} **Quality Audit Mechanism:** {{Centralized viral load monitoring and clinical site adherence tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{144 Follow-up weeks}} **Onsite Visit Cadence:** {{Regular intervals up to Weeks 48, 96, and 144}} **Remote Monitoring Frequency:** {{Continuous safety monitoring and AE tracking}} **Checklist Review Interval:** {{Every 12 weeks for adherence reviews}}

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					